

Mohamed Salama

Senior Software Engineer | Tech Lead | Distributed Systems & FinTech

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PROFESSIONAL SUMMARY

Senior Software Engineer and Tech Lead with **8+ years** architecting mission-critical systems across FinTech, high-frequency trading, and automotive verticals. Led distributed engineering teams of **12+ developers**, conducted technical hiring, and mentored interns while delivering measurable system performance and reliability improvements. Currently architecting and optimizing full-spectrum banking payment infrastructure – covering retail, corporate, and regulatory payment operations – in a regulated EU financial institution. Expert in Node.js, Java, gRPC, Kubernetes, and distributed systems. Multilingual: English, French, Arabic, German. Entrepreneurial background in digital product development.

SKILLS

Leadership & Management Technical Leadership (12+ devs), Engineering Team Lead, Technical Hiring, Intern Mentoring, Agile / SCRUM, Code Review Standards, Runbook Authoring, Cross-Functional Collaboration

System Architecture Distributed Systems, Microservices, Event-Driven Architecture, High Availability Design, Scalability Patterns, Circuit Breakers, API Gateway, Service Mesh

Backend Engineering Node.js (NestJS / Express), Java (OOP, Design Patterns), Python, TypeScript, REST / GraphQL APIs, gRPC / Protobuf, WebSockets, RabbitMQ, Kafka

Cloud & DevOps AWS (EC2, ECS, RDS, S3, Lambda), GCP (GKE, Compute Engine), Kubernetes, Docker, Terraform (Production), CI/CD (GitLab, GitHub Actions), Nginx

Observability & SRE Dynatrace, Elasticsearch, AWS CloudWatch, Datadog, ELK Stack, Distributed Tracing, Incident Management, Performance Optimization

Data & Storage PostgreSQL, MySQL, MSSQL, MongoDB, ArangoDB, Redis, SQL Optimization, Query Performance Tuning, Caching Strategies

FinTech & Banking Payment Systems Architecture, Standing Orders, Instant Payments, SWIFT, SEPA, Direct Debits, Banking Compliance, Payment Gateway Integration, Regulatory Financial Systems

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Payment Systems

Oct 2024 – Present

Keytrade Bank

Brussels, Belgium

- Led payment systems engineering for a regulated EU financial institution, architecting full-spectrum retail and corporate banking payment infrastructure including Standing Orders, Instant Payments, Memo Payments, SWIFT transfers, and SEPA Direct Debits.
- Delivered **50% performance improvement** by re-engineering standing order execution from legacy SQL-based jobs to an event-driven cronjob architecture with exponential backoff retry logic, reducing system load and enabling horizontal scaling.
- Designed and implemented independently a circuit-breaker “skip-record” mechanism that improved payment pipeline success rate from **92% to 99%**, preventing single transaction failures from cascading and blocking entire batch processing queues.
- Built observability platform replacing basic database logging with self-triggered success metrics and real-time anomaly detection using Dynatrace and Elasticsearch, significantly reducing payment failure detection and response time.
- Reduced new engineer onboarding time by **30%** through establishment of runbooks, architecture documentation, and structured mentoring protocols.
- Applied Infrastructure as Code using Terraform in production for cloud infrastructure management on AWS.

Stack: Node.js · React.js · RabbitMQ · Kafka · gRPC · Protobuf · AWS · PostgreSQL · Redis · Terraform · Dynatrace · Elasticsearch

Software Engineer, HFT Analytics Platform

Jul 2023 – Sep 2024

Features Analytics

Nivelles, Belgium

- Designed full-stack trading analytics platform bridging high-throughput computational backend with intuitive Vue.js dashboards, delivering actionable market intelligence to institutional clients.
- Optimised AI model analysis pipeline using decision tree algorithms, enabling near real-time processing of trade data for high-frequency analytics workloads.

Stack: Java (OOP) · Python · Node.js · Vue.js · PostgreSQL · AWS

Full-Stack Software Engineer & Team Lead

Sep 2022 – Jul 2023

SyIndr

Cairo, Egypt

VC-backed (\$12.6M pre-seed) automotive FinTech marketplace — one of Africa's largest pre-seed rounds — processing \$20–25K average vehicle transactions, achieving ~10× sales growth during tenure.

- Led distributed engineering team of **12+ developers** across backend, frontend, and mobile, establishing code review standards, CI/CD pipelines (GitLab), and agile sprint rituals.
- Architected secure, highly available multi-cloud microservices platform (AWS + GCP) integrating **15+ third-party APIs** including Paymob (payments), Cequens/Twilio (SMS communications), and a custom scheduling system for vehicle inspection appointments.
- Deployed production Kubernetes clusters with automated CI/CD release pipelines, eliminating manual deployment processes and enabling a consistent release cadence.
- Designed and built custom vehicle inspection scheduling and calendar system, coordinating warehouse visits and home delivery logistics for buyers.
- Conducted technical hiring interviews and mentored interns, contributing to team growth during a period of approximately tenfold sales growth.

Stack: Node.js · NestJS · React.js · React Native · PostgreSQL · Redis · MongoDB · Kubernetes · AWS · GCP · GitLab CI

Full-Stack Software Engineer

Apr 2021 – Sep 2022

Epicore

Cairo, Egypt

- Built 3 full-stack e-commerce applications with monolithic backend architecture, delivering end-to-end features in an agile environment.
- Designed and implemented scalable REST and GraphQL APIs applying industry best practices for API design, query optimisation, and database performance.
- Applied performance tuning techniques across PostgreSQL and MongoDB to improve response times and data throughput.

Stack: Node.js · Express · GraphQL · PostgreSQL · MongoDB

Full-Stack Developer

Jun 2019 – Mar 2020

LocusKey

Cairo, Egypt

- Architected backend server infrastructure integrating payment gateways, push notification services, and third-party booking APIs for an early-stage tourism platform.
- Designed and managed deployment pipeline, ensuring reliable release processes for platform services.
- Mentored junior developers on backend architecture, API design, and deployment best practices.

Stack: Node.js · PostgreSQL · Vue.js · AWS (EC2, RDS)

EDUCATION

Bachelor of Computer Science & Engineering

Sep 2016 – Jul 2021

German University in Cairo

Cairo, Egypt

R&D Engineer (Bachelor Thesis)

Mar 2020 – Aug 2020

DHBW Stuttgart

Stuttgart, Germany

- Developed an intelligent athlete performance tracking system using Computer Vision and Machine Learning to optimize cyclist training behavior.

Stack: Python · OpenCV · NumPy · Pandas · Keras

CERTIFICATIONS

AWS Certified Cloud Practitioner · Issued July 2023, valid through July 2026. Currently advancing toward next AWS certification tier.

LANGUAGES

Arabic Native · **English** Fluent · **French** Conversational · **German** Basic

ENTREPRENEURIAL VENTURES

- **Travel Guide & Planner** — Digital travel planning product.
- **E-Commerce Platform** — Early-stage experimental e-commerce platform.

PERSONAL PROJECTS

- **Rental Management App** (TypeScript) — A versatile full-stack property rental management application for landlords and property owners.
- **MovieTrailerEngine** (Vue.js / Node.js) — Distributed movie discovery platform handling concurrent users with microservices architecture.
- **Apartment Search Engine** (JavaScript) — Property search and filtering platform.